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Influence of internal heating and natural convection on airborne transmission of viral diseases

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Abstract

We conduct large eddy simulations in a canonical room of size $10 \times 10 \times 3.2\,\mathrm{m}^3$ subject to mechanical ventilation to assess the impact of internal heating on droplet nuclei dispersal in indoor spaces. We conduct two simulations with a solitary heat source placed at the center of the room in the first, and 16 heat sources arranged in an equidistant non-staggered formation in the second. The heat sources mimic heat generation from an adult person. The simulations employ a statistical overloading approach where 10 million droplet nuclei are individually monitored for approximately one hour in real-time. These results are then compared to previous simulations that share identical setup and procedures but do not account for any thermal load. We assess the effect of internal heat sources on multiple levels by averaging over all possible positions of the infected (source) and the susceptible (sink) occupants within the room. We find that, at the room-average level, the room concentration, which results from a balance of droplet nuclei production through expiratory events and removal through various mechanisms is unaffected by the presence of heat sources. On the other hand, the spatial and temporal dispersion of droplet nuclei, as

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observed from the standpoint of a known source or sink location, or when a combination of sources and sinks at fixed separation distances is considered, is significantly influenced by the presence of heat sources.

Keywords: large eddy simulations, internal heating, indoor viral spreading, well-mixed theory

1. Introduction

Airborne diseases within enclosed environments and overall indoor air quality have garnered substantial focus, particularly in the recent years following the COVID-19 pandemic (Liu *et al* 2020, Zhang *et al* 2020, Foster and Kinzel 2021, Schimmoller 2022). This subject is undeniably interdisciplinary, with a significant emphasis on understanding the fluid dynamics of the dispersion of micron-sized droplet nuclei within confined areas (Ching *et al* 2009, Abkarian and Stone 2020, Vuorinen 2020, Mathai *et al* 2021, Wang *et al* 2021, Fabregat *et al* 2021a, 2021b). These nuclei may carry viruses and can become suspended within the indoor environment as a result of expiratory activities like coughing, sneezing, singing, talking, breathing, etc. Elevated concentrations of these particles within indoor spaces typically correspond to an increased risk of infection. Hence, it is imperative to acquire accurate estimates of the spatial and temporal dispersal patterns of emitted nuclei within enclosed areas (Memarzadeh and Xu 2012, Motamedi Zoka *et al* 2021, Motamedi *et al* 2022, Salinas *et al* 2022). This however, continues to be a challenging endeavor owing to the chaotic nature of turbulence, as well as potential uncertainties associated with the location and quantity of infected individuals, as well as the location of potential recipients.

In order to address these challenges, simple yet insightful models, such as the well-mixed model (Wells *et al* 1955, Riley *et al* 1978, Bazant and Bush 2021), have been applied within the context of airborne contagion. These models aim to estimate the concentration of particles, and consequently the risk of infection, by additionally taking into consideration viral properties and physiological factors (Buonanno *et al* 2020, Sun and Zhai 2020, Bazant and Bush 2021, Miller *et al* 2021). From a fluid mechanics point of view, the well-mixed model operates under the assumption that the ejected droplet nuclei instantly and uniformly disperse throughout the indoor area. Consequently, the infection risk calculated using the well-mixed model is independent of a crucial factor: the distance between the infected individual (referred to as the 'source') and the potential host (referred to as the 'sink'). In other words, according to the well-mixed model, an individual's risk of infection remains the same, whether they are situated 2 meters or 10 meters away from an infected person. This deficiency of the well-mixed model has been addressed with the use of near/far-field model (Nicas 1996, 2016) and eddy diffusivity model with wall reflections (Drivas *et al* 1996, Nicas *et al* 2009, Venkatram and Weil 2021).

Recently, a statistical overloading approach that aims at providing statistically relevant estimates for the spatio-temporal distribution of droplet nuclei was proposed using large eddy (Salinas *et al* 2022, Krishnaprasad *et al* 2023) as well as Reynolds-averaged Navier–Stokes simulations (Choudhary *et al* 2023a, 2023b). The statistical overloading involves tracking millions of droplet nuclei, far in excess of what would be present in a typical indoor space. One of the main outcomes was the development of a source–sink, distance-dependent correction function to be used in conjunction with the well-mixed model. The correction function provided a multiplicative factor that would account for the time the source and the sink spend in the indoor space as well as their separation distance.

Within the context of indoor airborne contagion, we identify two aspects of the flow that strongly affect the risk of viral transmission. The first is the rate at which fresh air enters the room which under steady state equals the rate at which contaminated air leaves. This rate of air exchange measured in air changes per hour (ACH) along with the rate at which airborne droplet nuclei are ejected into the room by the source determines the room-averaged nuclei concentration. The second is the degree of turbulent mixing within the room. In a room that is well mixed, the local nuclei concentration at different points within the room will not substantially deviate from the room-averaged value. On the other hand, in a room that is not well mixed, there will be regions of substantially higher or lower nuclei concentration than the room-averaged concentration. In a typical force-ventilated room, the air-conditioning (AC) unit determines both the ACH as well as the degree of turbulent mixing within the room. The two are usually well-correlated in the sense that a large ACH usually indicates strong turbulent mixing. The same is perhaps true of a room that is naturally ventilated in an efficient manner. However, in certain extreme situations, such as a sealed room with a fan, the ACH may be low despite significant mixing.

In this regard, internal heating within the room resulting from human bodies or electronic gadgets can play an important role. In the case of natural ventilation, such internal heat sources determine both the ACH as well as the level of local mixing. In the case of forced ventilation, the local heat sources may not alter the ACH dictated by the AC unit but may alter the local level of turbulent mixing. In essence, the airflow patterns within the room solely driven by mechanical ventilation can be substantially altered by the added convection currents induced by the heat sources.

In studying the fluid mechanics of airborne viral transmission in confined spaces, a common approach involves releasing particles of various sizes to simulate droplet nuclei from expiratory events at a specific location and rate. Sensors are then placed at different points to measure particle concentrations. This setup is typically used to assess the risk of infection for susceptible individuals positioned at the sink (sensor location) when a sick person is located at the particle source. Consequently, assessing the risk of infection from a different particle source requires recomputation with appropriate source and sink positions.

The statistical overloading technique presented in this paper greatly simplifies the approach. We employ a single simulation that is capable of assessing the risk of infection from any source location to sink location, for any ejection rate and particle size spectra. This is achieved by uniformly and randomly distributing over 10 million droplet nuclei of various sizes throughout the indoor space. We track the trajectory of all the individual droplet nuclei over the course of an hour. This detailed Lagrangian trajectory database contains information on droplet nuclei that travels from any source location within the room to any sink (or sensor) location within the room, which allows calculation of the probability of droplet nuclei reaching a sink after being ejected from a source, for every source/sink combination within the room. This statistical overloading framework was initially proposed by Salinas *et al* (2022), and has been implemented subsequent studies (Krishnaprasad *et al* 2023, 2024, Choudhary *et al* 2023a, 2023b). However, these studies did not consider the effect of natural convection brought on by the presence of heat sources in the confined space. The statistical overloading of the room with many millions of droplet nuclei (far more than in any one actual scenario) was possible due to the one-way coupled nature of the flow, where the ejected droplets do not alter the room-scale flow.

The objective of the present work is to investigate the effect of heat sources on airborne transmission of viral diseases (Wang and Chen 2007, Foat *et al* 2017, Sun *et al* 2021). More specifically, we will build on the work of Salinas *et al* (2022), Krishnaprasad *et al* (2023),

Choudhary *et al* (2023a) by adding heat sources to the indoor space. The heat sources will modify the air temperature and consequently its density thereby generating a buoyancy-driven flow. We are interested in exploring how the added flow turbulence and mixing brought on by the heat sources would modify the risk of indoor airborne contagion by modifying the concentration of pathogen-carrying droplet nuclei within the room. The concentration is directly affected by nuclei trajectories, which are primarily shaped by the modified air-flow patterns.

We will assess the impact across four distinct tiers: (i) The average concentration of nuclei within the room. In the context of mechanical ventilation, droplet nuclei remain well-mixed within the room. This average room concentration depends only on the balance between the production of droplet nuclei from an expiratory source and the decay rate of these nuclei through ventilation, turbulent wall deposition, and floor settling. As such, the temporal evolution of the average concentration of droplet nuclei within the room is independent of where the source and sink are located within the room. (ii) The nuclei concentration from the perspective of the source (sick person) location: This situation reflects the scenario where the source's location is known, but the location of the receiving host is a variable. Here, the nuclei concentration has been averaged over all possible sink locations given the source location. (iii) The nuclei concentration from the perspective of the sink (receiving host) location: This situation reflects the scenario where the receiving host's location is known, but the location of the source is uncertain. Thus, the nuclei concentration at the sink location has been averaged over all possible source locations. (iv) The nuclei concentration when the source and sink are separated by a known distance d, yet their precise locations are both uncertain. This scenario is relevant for social distancing of say one or two meters apart. We would like to highlight that 'heat source' and 'source' are not used interchangeably, i.e. the heat source is not a source of virus-laden droplets.

2. Numerical methodology

Our domain consists of a canonical room of size $10\,\mathrm{m} \times 10\,\mathrm{m} \times 3.2\,\mathrm{m}$ with a four-way cassette AC unit placed at the center of the ceiling. The unit is composed of 4 inlets of size $5.5\,\mathrm{cm} \times 44\,\mathrm{cm}$ surrounding a central outlet of size $60\,\mathrm{cm} \times 60\,\mathrm{cm}$. The vane angle of the unit is set at 40° relative to the ceiling (see Inset (D) in figure 1). Note that the inlet and outlet surfaces are perfectly aligned with the ceiling (i.e. they do not protrude into the room.) Two simulations are considered where volumetric heat sources of radius $R = 29\,\mathrm{cm}$ and height $L = 1.6\,\mathrm{m}$ are positioned in contact with the floor. In the first simulation, a solitary heat source is placed at the center of the room, whereas in the second simulation, 16 sources are placed equidistantly in a non-staggered arrangement as shown in panels (a) and (b) of figure 1. The sources emit heat at the rate of $E = 168.8\,\mathrm{W}$ mimicking the heat generated by an average adult (ASHRAE 1993, Murakami *et al* 2000). The fluid is air with density $\rho_\mathrm{f} = 1.2\ \mathrm{kg\,m^{-3}}$ and dynamic viscosity $\mu = 1.81 \times 10^{-5}\,\mathrm{Pa \cdot s}$. The air is ejected from the inlets at a temperature of $T_\mathrm{in} = 289\,\mathrm{K}$ with a velocity of $U_\mathrm{in} = 3.57\,\mathrm{m\,s^{-1}}$, which results in an inlet Reynolds number of $\mathrm{Re_{in}} = 4\rho_\mathrm{f} U_\mathrm{in}\sqrt{A_\mathrm{in}}/\mu_\mathrm{f} = 1.47 \times 10^5$, where $A_\mathrm{in} = 2.42 \times 10^{-2}\,\mathrm{m^2}$ represents the surface area of an individual inlet. The air inlet velocity is chosen such as to achieve an ACH rate of 2.5. The inlet jet velocity is strong enough to prevent short-circuiting of flow and droplet trajectories from the inlet to the ventilation outlet.

We solve the conservation equations of mass (1), momentum (2), and energy (3) through large eddy simulations (LES) (Boris *et al* 1992) using the dynamic Smagorinsky model (Germano *et al* 1991, Lilly 1992, Balachandar *et al* 2019):

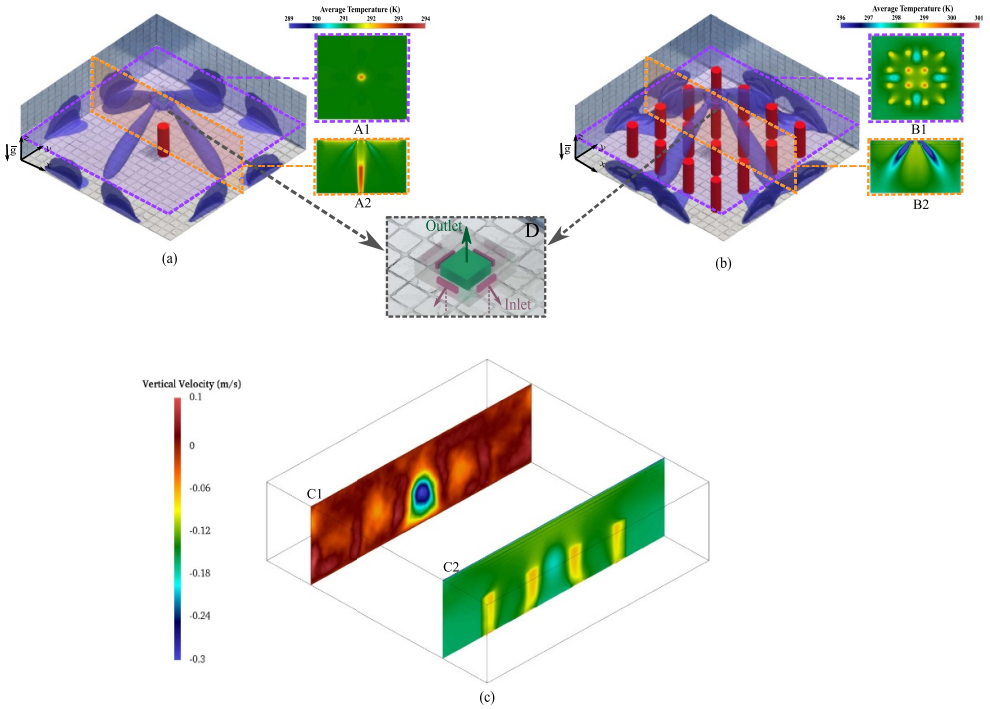


Figure 1. Isocontours of velocity magnitude of $0.2\,\mathrm{m\,s^{-1}}$ in the main frame along with the location of the heat source(s) (red cylinders) for (a) a solitary heat source and (b) 16 heat sources in a non-staggered configuration. The insets (A) and (B) show temperature profiles along the purple ($z = 1.6\,\mathrm{m}$) and orange ($y = 5\,\mathrm{m}$) planes. Inset (D) shows the air-conditioning unit setup. (c) shows the case with 16 sources, where (C1) is a vertical slice taken at ($x = 2\,\mathrm{m}$) , depicting the profile of average vertical velocity. (C2) is another vertical slice taken at ($x = 8\,\mathrm{m}$), illustrating the profile of average temperature.

$$\nabla \cdot \mathbf{u} = 0 \tag{1}$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\,\mathbf{u} = -\frac{1}{\rho}\nabla p + (\nu + \nu_{\mathrm{t}})\,\nabla^2 \mathbf{u} - \mathbf{g}\beta\,(T - T_0) \tag{2}$$

$$\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T = \left(\frac{k}{\rho C_p} + \frac{\nu_t}{Pr}\right)\nabla^2 T + [P]\,\frac{E}{\rho C_p}. \tag{3}$$

Here, $\mathbf{u}, p$, and T represent the velocity, pressure, and temperature fields, respectively. Additionally, $\nu = 1.5 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$ represents the kinematic viscosity of air, $C_p = 1005\,\mathrm{J\,kg^{-1}\,K^{-1}}$ the specific heat capacity, $k = 0.024\,\mathrm{W\,m^{-1}\,K^{-1}}$ the thermal conductivity, $\beta = 3.43 \times 10^{-3}\,\mathrm{K^{-1}}$ the coefficient of thermal expansion, $\mathbf{g} = (0, 0, 9.81)\,\mathrm{m\,s^{-2}}$ the gravitational acceleration and ν_{t} the turbulent eddy viscosity. $T_0 = 293\,\mathrm{K}$ is a reference temperature, and $[P]$ is a masking function such that $[P] = 1$ inside the heat source and 0 elsewhere.

As far as boundary conditions are concerned, we impose zero velocity on all surfaces of the room except at the inlet and outlet. As previously mentioned, the inlet velocity is $3.57\,\mathrm{m\,s^{-1}}$ oriented at an angle of 40° with respect to the ceiling, whereas an open boundary condition with zero pressure is imposed at the outlet. As far as temperature is concerned, we impose a zero-flux adiabatic condition on all surfaces with the exception of the inlet and outlet. The

incoming air is maintained at 289 K, and the room air is allowed to freely exit at the outlet. For the present configuration, after the flow reaches a statistically-steady state, we observe the average temperature in the room with the solitary heat source to reach 291 K. On the other hand, the room with the 16 heat sources reaches an average temperature of 298 K.

As can be seen from slice (C2) in figure 1(c), there are significant spatial variations of temperature within the room. We note that the air temperature within the heated cylindrical sources are about 2 K higher compared to the room average. This is true for both the single source and the multiple source configurations. Additionally, in panel (C1), we show the vertical component of velocity in a vertical slice at $x = 2$ m. We observe the velocity in the slice to vary in the range -0.3 to $0.1\,\mathrm{m\,s^{-1}}$. Moreover, we observe the influence of the heat sources on the flow whereby the air within the heat sources is directed upwards (dark red contours) whereas the air in between the heat sources is directed downwards (orange contours). The slice also cuts through the inlet jet where strong downward pointing air velocities are observed.

Once the flow reaches the aforementioned steady state, we follow the statistical overloading approach of Salinas *et al* (2022), Krishnaprasad *et al* (2023, 2024), Choudhary *et al* (2023a, 2023b) and seed the room with 10 million droplet nuclei of 10 different sizes. The nuclei sizes considered are $r = 0.1$, 0.5, 2.5, 5, 7.5, 10, 12.5, 15, 20, and 25 μm. The nuclei are modelled as point particles. They are advected by the flow and are individually tracked for the duration of the simulation. Additional details on the numerical methodology of the present Euler–Lagrange simulations can be found in Salinas *et al* (2022), Krishnaprasad *et al* (2023). It is also important to note that droplet evaporation typically occurs on a timescale on the order of one second. In contrast, the timescales of interest extend orders of magnitude beyond the evaporative timescale and as such are not accounted for. In other words, the droplet nuclei considered here have already reached their terminal size and are devoid of any further evaporation.

To clarify the implementation of the present statistical overloading framework, we describe how every possible source–sink configuration within the room is simultaneously addressed as part of post-processing. We discretize the room into $25 \times 25 \times 8$ virtual spheres of diameter $D_s = 0.4$ m. Each of these spheres can represent both a source and a sink volume and as a result, we address approximately 25 million source–sink combinations. As millions of droplet nuclei are tracked within the room for about an hour, we typically observe $O(10^4)$ to $O(10^5)$ of these nuclei to travel from any source sphere volume to any other sink volume, thus providing ample sample size for accurate calculation of droplet nuclei statistics. In particular, for any chosen source and sink volumes, we monitor all the droplet nuclei that were observed within the source volume and investigate the fraction of these droplet nuclei that reach the sink volume at a later time and the associated travel time. This framework allows accurate calculation of nuclei concentration at the sink volume for all possible source–sink locations with a single simulation.

The presence of heat sources can be regarded as an additional thermal ACH acting in the room and causing localized convection. The volume flow rate Q, into and out of the room, for an ACH of 2.5, is calculated to be about $0.222\,\mathrm{m^3\,s^{-1}}$. Due to the heat input, a temperature difference is produced within the domain as

$$\Delta T = T_{\mathrm{out}} - T_{\mathrm{in}} = \frac{E}{\rho C_p Q}, \tag{4}$$

where T_{out} is the temperature at the outlet. For a single heat source, this temperature difference comes to about 0.63 K.

3. Results and discussion

3.1. Room-averaged statistics

The impact of internal heating is first assessed by examining the temporal variation of room-averaged nuclei concentration, where the averaging is over all possible source and sink locations. In the absence of heating, but in the context of mechanical ventilation, the room-averaged concentration depends only on the balance between the expiratory event and the decay rate of nuclei through ventilation, turbulent wall deposition, and floor settling (Bazant and Bush 2021, Salinas *et al* 2022, Krishnaprasad *et al* 2023, Choudhary *et al* 2023a). Based on this balance, the well-mixed theory predicts an exponential approach to a constant steady-state concentration after the arrival of the source into the room, whereby the exponential rate of approach to steady state depends on nuclei size (Bazant and Bush 2021).

For an indoor space of volume $\mathscr{V}$, the time-dependent normalized nuclei concentration for nuclei of radius r may be expressed as (Wells *et al* 1955, Riley *et al* 1978, Bazant and Bush 2021, Miller *et al* 2021, Salinas *et al* 2022)

$$\hat{n}_{wm}(t,r) = \frac{1}{\lambda_{wm}(r)\mathscr{V}}\left(1-\exp\left(-\lambda_{wm}(r)\,t\right)\right), \tag{5}$$

where the exponential decay rate of the nuclei is given by

$$\lambda_{wm}(r) = \frac{\mathrm{ACH}_{\mathrm{ef}}}{3600} + \frac{V_{\mathrm{s}}(r)}{H}. \tag{6}$$

Here $V_{\mathrm{s}}(r)$ and H represent the size-dependent droplet settling velocity and room height, respectively. The viral deactivation rate is taken to be zero in the present case.

$\mathrm{ACH}_{\mathrm{ef}}$ in (6) represents the effective ACH that accounts for recycling of air and filtration (Krishnaprasad *et al* 2023). The LES results are used to calculate the normalized nuclei concentration $\hat{n}(t,\mathbf{x}_{\mathrm{so}},\mathbf{x}_{\mathrm{si}},r)$, which, unlike the well-mixed approximation, is now also a function of the source and sink locations. If we perform a double average over all possible source and sink locations, we obtain the room-averaged normalized nuclei concentration as $\langle\langle\hat{n}(t,r)\rangle\rangle = \iint \hat{n}(t,\mathbf{x}_{\mathrm{so}},\mathbf{x}_{\mathrm{si}},r)\mathrm{d}\mathbf{x}_{\mathrm{so}}\mathrm{d}\mathbf{x}_{\mathrm{si}}/\mathscr{V}^2$, where the integrals are over the entire room volume. The present LES results $\langle\langle\hat{n}(t,r)\rangle\rangle$ closely followed the exponential rate given in (5) with a relative error on the order of 1% for the smallest micron-sized nuclei. The numerical decay rates are then obtained by fitting these exponential trends.

In figure 2(a), we plot the decay rate λ vs nuclei radius r predicted by the well-mixed theory as given in (6) along with values obtained from the exponential fit of three LES results: (i) without any heat source, (ii) with a single heat source, and (iii) with 16 heat sources. For the three simulations considered, λ represents the total decay rate due to ventilation, floor settling, and wall deposition. Additionally, the cross symbols in the figure correspond to the theoretical estimate of λ from the well-mixed approximation. A small value of λ indicates that, on average, nuclei will remain suspended in the room for a relatively long time following their ejection and correspondingly a large value of λ indicates faster removal of nuclei from the room. Figure 2(b) shows the fraction of particles that exited the room by settling on the floor, through the outlet, or by depositing on the walls, represented as $\tilde{\chi}_{\mathrm{f}}$, $\tilde{\chi}_{\mathrm{o}}$, and $\tilde{\chi}_{\mathrm{w}}$, respectively.

From figure 2(a) it is clear that for droplet nuclei of radius smaller than 5 μm, the effect of heat sources is minimal and all three simulation results are in good agreement with the well-mixed theory. Furthermore, these smaller nuclei are primarily removed through the ventilation outlet. Figure 2(b) shows that the presence of heat sources has little to no effect on the mode

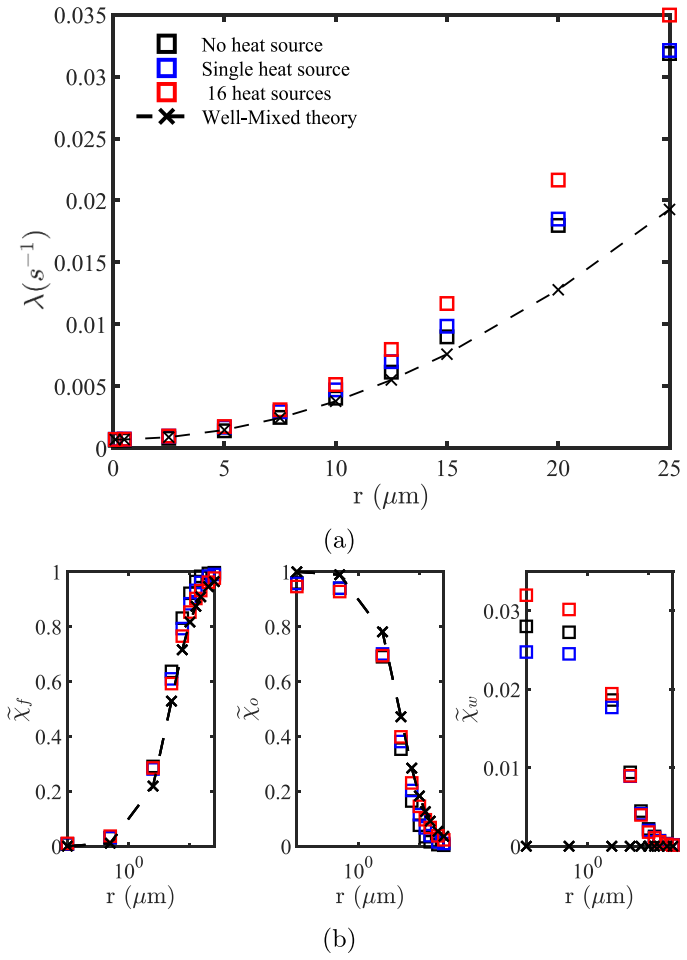


Figure 2. (a) Decay rate λ vs nuclei radius r from the LES results with no heat sources (Salinas *et al* 2022) (black squares), with a single heat source (blue squares), with multiple heat sources (red squares), and from the well-mixed model (dashed line with black crosses). (b) The fraction of nuclei that are removed from the room by settling to the floor($\tilde{\chi}_f$), through the ventilation outlet($\tilde{\chi}_o$), or by depositing on the walls and ceiling($\tilde{\chi}_w$).

of particle removal as well. The excellent agreement between the well-mixed theory and the LES results for micron-sized nuclei indicates that nuclei remain well-mixed in the presence of internal heating. The largest disparity observed between the simulation and the well-mixed results for $\tilde{\chi}_w$ is the result of the fact that the well-mixed theory does not account for wall deposition. The theory also overpredicts $\tilde{\chi}_o$ by about 5%–15%.

The overall removal rate of larger nuclei is under-predicted by the well-mixed theory. These large nuclei are primarily removed by settling on the floor, with only a small fraction exiting through the ventilation outlet or depositing on the walls and roof (figure 2(b)). The presence of heat sources systematically increases the decay rate, with 16 internal heat sources enhancing nuclei removal for the largest size by about 10% compared to the no-heat-source case. Enhanced floor deposition is observed for the cases with heat sources, particularly at the initial

stages. This can be attributed to the lower air density within the heat sources, which causes air to rise, reducing deposition within those regions. In contrast, cooler temperatures outside the heat sources push air downward, increasing floor deposition. Thus, the likely cause of the increased deposition is that the downwelling regions outside the heat sources are larger than the upwelling regions inside, leading to a net increase in floor deposition. Additionally, the upwelling air helps keep larger particles airborne for a longer period, increasing their likelihood of being removed via the ventilation outlet.

3.2. Sink-averaged source perspective

Next, we proceed to assess the impact of internal heating from a source's perspective, i.e. from the perspective of an infected individual, by averaging over all possible sink locations. In other words, we consider the scenario where the sink location is a variable and compute the steady-state room-average concentration for various nuclei sizes depending on the location of the source. Specifically, we are interested in how the presence of internal heat sources affects the long-term, spatio-temporal distribution of nuclei concentration. Figure 3(a) shows contours of the normalized nuclei concentration $\mathscr{N}_{\mathrm{so}}$ in the horizontal $z = 1.6$ m plane (purple plane shown in figure 1) bisecting the middle of the room. The A, B, and C frames correspond to the case with no heat sources, the case with the solitary heat source, and the case with the multiple heat sources, respectively. Following Salinas *et al* (2022), $\mathscr{N}_{\mathrm{so}}$ is defined as

$$\mathscr{N}_{\mathrm{so}}\left(\mathbf{x}_{\mathrm{so}}, r\right) = \frac{\left(\int \hat{n}\left(t \to \infty, \mathbf{x}_{\mathrm{so}}, \mathbf{x}_{\mathrm{si}}, r\right) \mathrm{d}\mathbf{x}_{\mathrm{si}}\right) / \mathscr{V}}{\langle\langle \hat{n} \rangle\rangle \left(t \to \infty, r\right)}. \tag{7}$$

The definition of $\mathscr{N}_{\mathrm{so}}$ in (7) corresponds to the long-time, steady-state concentration in the room for a particular source location ($\mathbf{x}_{\mathrm{so}}$), averaged over all possible sink locations ($\mathbf{x}_{\mathrm{si}}$), and normalized by the room-average concentration ($\langle\langle \hat{n} \rangle\rangle$) at steady-state (i.e. as $t \to \infty$).

We note here that while in the simulations, we release a large number of nuclei at the start of the simulation from which we compute the decay rate for that instantaneous release, in equation (7) we compute the long-term average concentration for a continuous release. In other words, equation (7) corresponds to a step function response to a continuous ejection, which is derived by integrating the delta function response for an instantaneous ejection. The iso-contours are shown for 3 nuclei sizes, namely $r = 0.1$, 7.5, and 25 μm.

The contour plots can be understood as follows: The PDFs are colored by the values of $\mathscr{N}_{\mathrm{so}}$, the mean value of which is 1 by definition. Regions with $\mathscr{N}_{\mathrm{so}} < 1$, therefore, represent preferable positions to locate the source of infection, since the corresponding steady-state nuclei concentration is lower than the well-mixed average. Differently put, at each location on this plot, the color corresponds to the increase in the room averaged concentration above the value obtained from the well-mixed model. For example, the contour plot of panels (B1) (single heat source) and (C1) (16 heat sources) indicate that when the source is positioned in the blue region, i.e. below the outlet, the concentration of sub-micron nuclei ejected by that source is lower, when averaged over all possible sink locations within the room. Alternatively, when the source is located in the dark red regions, then the sink-averaged nuclei concentration can be expected to be larger. The behavior of nuclei of size $r = 7.5\,\mu$m in panels (A2), (B2), and (C2) are somewhat different, although the optimal location for the source still remains directly below the central outlet. For even larger nuclei of size $r = 25\,\mu$m, the blue regions (i.e. regions with the smallest values of $\mathscr{N}_{\mathrm{so}}$) lie in the path of the inlet jets where such large nuclei are pushed towards the floor thereby enhancing their gravitational settling rate. Due to the generally overwhelming abundance of sub-micron droplet nuclei, we observe that the safest place

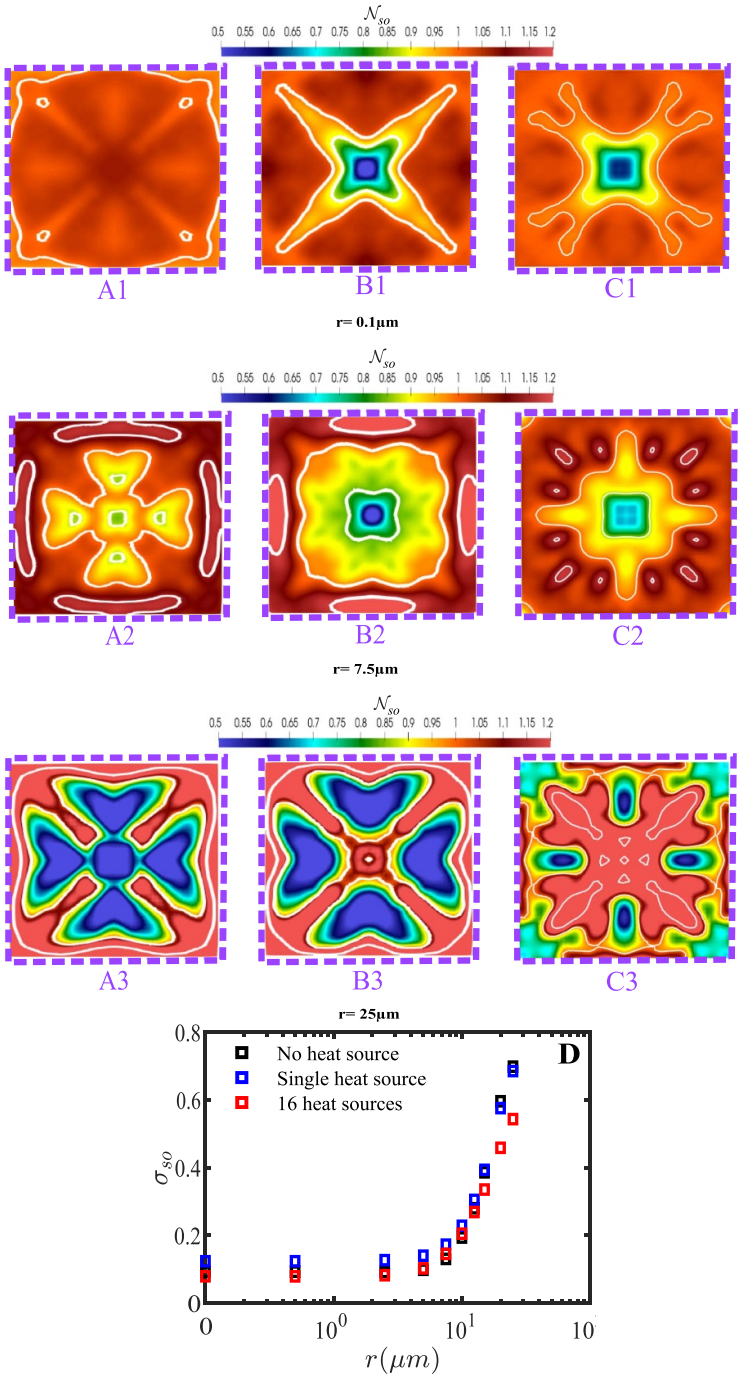


Figure 3. Concentration from the source's perspective for ACH = 2.5. (A), (B), and (C) show contours of nuclei concentration $\mathcal{N}_{so}$ for $r = 0.1$, 7.5, and 25 μm at $z = 1.6$ m for the cases with no heat sources, a single heat source and multiple heat sources, respectively. (D) shows the standard deviation of the PDFs of $\mathcal{N}_{so}$ for the entire room as a function of nuclei radii.

to position an infection source is immediately below the outlet such that a large portion of these nuclei are directly removed by ventilation and not circulated within the room. Clearly, this is not a surprising result, which can be expected based on simple reasoning. Nevertheless, the LES results provide a quantitative measure that the reduction in the concentration of small nuclei is about 18%. We also find that the above observation is true for nuclei of size up to $r = 7.5\,\mu$m, i.e. for nuclei whose settling velocity is smaller than the upward airflow directed towards the AC outlet, allowing these nuclei to be carried by the air flow towards the outlet, and thereby removing them from circulation.

To assess the impact of the solitary heat source, we compare the contour plots in the (B) panels against the corresponding contour plots of Salinas *et al* (2022) shown in the (A) panels. We observe the contours pertaining to the solitary heat source to be substantially different than those with no heat source as a result of the altered flow field due to the presence of the heat source. In the case of the sub-micron-sized nuclei, we find that in the presence of a heat source directly positioned below the ventilation outlet, a region of low nuclei concentration (blue iso-contour) is created as a result of the rising air caused by the elevated temperatures near the heat source. This buoyancy-driven flow is absent without the heat source, as evident by the presence of a red region in panel (A1) of figure 3. Similar behavior is observed for the $r = 7.5\,\mu$m nuclei. For the case of the larger droplet nuclei of $r = 25\,\mu$m, the upwelling of airflow now acts to partially counterbalance the downward motion of the relatively large nuclei, thereby allowing them to be advected by the indoor air currents and consequently increase their residence time in the room.

For the largest nuclei of $r = 25\,\mu$m, we still observe from panel (C3) that the smallest values of $\mathscr{N}_{\text{so}}$ occur in the path of the inlet jets where the downward air flow acts to enhance the gravitational settling of these relatively large nuclei. While in the solitary heat source case, we observe the outer edges of the room to reflect the largest values of $\mathscr{N}_{\text{so}}$, in the presence of many heat sources, the convective flow acts to enhance mixing in these areas and thereby reduces the concentration of larger nuclei near the walls.

We recall that the mean value of nuclei concentration is given by equation (5) with the decay rates given in figure 2(a). The level of concentration variation about this mean value shown in the contours is quantified in terms of standard deviation of nuclei concentration σ_{so}, which is shown in panel (D) of figure 3 presented against the nuclei radius for all three cases. A larger value of σ_{so} indicates an inhomogeneous distribution of nuclei concentration about the mean and significant departure from the assumption of well-mixedeness. From the figure, it is clear that in the present case, uniformly distributed heat sources do not enhance or diminish mixing of the smallest size nuclei, where the value of σ_{so} from the no heat source case and multiple heat sources case are practically identical. This also suggests that the small nuclei are already well-mixed by the ventilation-driven flow. On the other hand, we do observe the heat sources to enhance mixing of larger nuclei as suggested by the smaller values of σ_{so}. For the case of the single heat source, we notice larger values of σ_{so} compared to the other two cases. This suggests a de-mixing effect for the single heat source case. This behavior is likely a result of the non-uniformity of heating as well as the placement of the heat source at the center of the room immediately below the ventilation outlet.

3.3. Source-averaged sink perspective

In this section, we assess the impact of internal heating on nuclei concentration from the perspective of different sink locations within the room by averaging over all possible source locations. Differently put, we consider the ideal locations for a potential receiving host (i.e. the sink) within the room when the source location is not precisely known. Specifically, we

are interested in how the presence of internal heat sources affects the long-term, spatio-temporal distribution of nuclei concentration at the sink location. Figure 4 shows iso-contours of the normalized nuclei concentration $\mathscr{N}_{\mathrm{si}}$ in panels (A)–(C) for the no heat source, solitary heat source, and 16 heat sources cases, respectively. Following Salinas *et al* (2022), $\mathscr{N}_{\mathrm{si}}$ is defined as

$$\mathscr{N}_{\mathrm{si}}\left(\mathbf{x}_{\mathrm{si}}, r\right) = \frac{\left(\int \hat{n}\left(t \to \infty, \mathbf{x}_{\mathrm{so}}, \mathbf{x}_{\mathrm{si}}, r\right) \mathrm{d}\mathbf{x}_{\mathrm{so}}\right) / \mathscr{V}}{\left\langle\left\langle \hat{n} \right\rangle\right\rangle\left(t \to \infty, r\right)}. \tag{8}$$

$\mathscr{N}_{\mathrm{si}}$ represents the long-time, steady-state concentration in the room for a particular sink location, averaged over all possible source locations, and normalized by the room average. In the panels, the iso-contours are shown at the $z = 1.6\,\mathrm{m}$ plane (purple horizontal plane in figure 1) for three nuclei sizes, namely $r = 0.1\,\mu\mathrm{m}$, $r = 7.5\,\mu\mathrm{m}$, and $r = 25\,\mu\mathrm{m}$.

The contour plots can be understood as follows: regions with $\mathscr{N}_{\mathrm{si}} < 1$ indicate locations where the risk of infection of a susceptible individual (i.e. sink) is low due to the lower concentration of virus-laden droplet nuclei when averaged over all possible source locations. The opposite applies to regions of $\mathscr{N}_{\mathrm{si}} > 1$. The level of concentration variation seen in the contours is quantified in terms of standard deviation σ_{si}, which is shown in panel (D) of figure 4 presented against the nuclei radius for all three cases.

Panels (A1), (B1), and (C1) show contours of $\mathscr{N}_{\mathrm{si}}$ for the smallest nuclei size of $r = 0.1\,\mu\mathrm{m}$. The results are however applicable for small nuclei of size $r \lessapprox 2.5\,\mu\mathrm{m}$. The panels show that the safest location for a susceptible person is in the direct path of the inlet jets. This is to be expected since we are considering perfectly clean incoming air that is devoid of any virus-carrying nuclei. On the other hand, we find that being located near the center of the room or along the diagonals exposes the receiver to the largest concentration of sub-micron and micron-sized nuclei. We should note however that the variation in $\mathscr{N}_{\mathrm{si}}$ remains small as inferred by the color map range indicating that these sub-micron nuclei are well-mixed within the room. Additionally, we do observe that the presence of the uniformly distributed heat sources in C1 shows a reduction in the range of $\mathscr{N}_{\mathrm{si}}$ indicating better mixing.

Panels (A2), (B2), and (C2) correspond to $\mathscr{N}_{\mathrm{si}}$ for nuclei size of $r = 7.5\,\mu\mathrm{m}$. The region of the lowest nuclei concentration still remains in the direct path of the inlet jets. However, while there was a region of low concentration directly below the outlet in the absence of any heat source as shown in panel (A2) of figure 4, in the presence of a single heat source directly below the outlet, upwelling of air in this region now tends to increase local nuclei concentration available to the sink. A large blue region is observed in panel (C2) implying the buoyancy-driven flow generated by multiple heat sources is strong enough to effectively mix intermediate-sized nuclei. The enhanced mixing in the multiple heat sources case is evident in the standard deviation plot in panel (D) as indicated by the generally lower values of σ_{si}. σ_{si} is observed to be the smallest compared to the no-heat and single heat source cases being on average about 30% lower than the aforementioned two cases.

Panels (A3), (B3) and (C3) correspond to $r = 25\,\mu\mathrm{m}$. These large nuclei have the largest range of $\mathscr{N}_{\mathrm{si}}$ values compared to other nuclei. For panels (A3) and (B3), the nuclei are concentrated in the corners of the room away from the direct path of the inlet jets. Due to the distribution of the 16 heat sources, the nuclei are further dispersed in the corners and get concentrated directly under the ventilation outlet.

The overall conclusion is that the risk of contagion is lower when the sink is located in the inlet jets of fresh outside air. Furthermore, this conclusion is not altered by the presence or absence of internal heat sources. We note here that the present simulations are in the limit of either no recycling of exhaust air back into the room or perfect filtration of any recycled air (Li

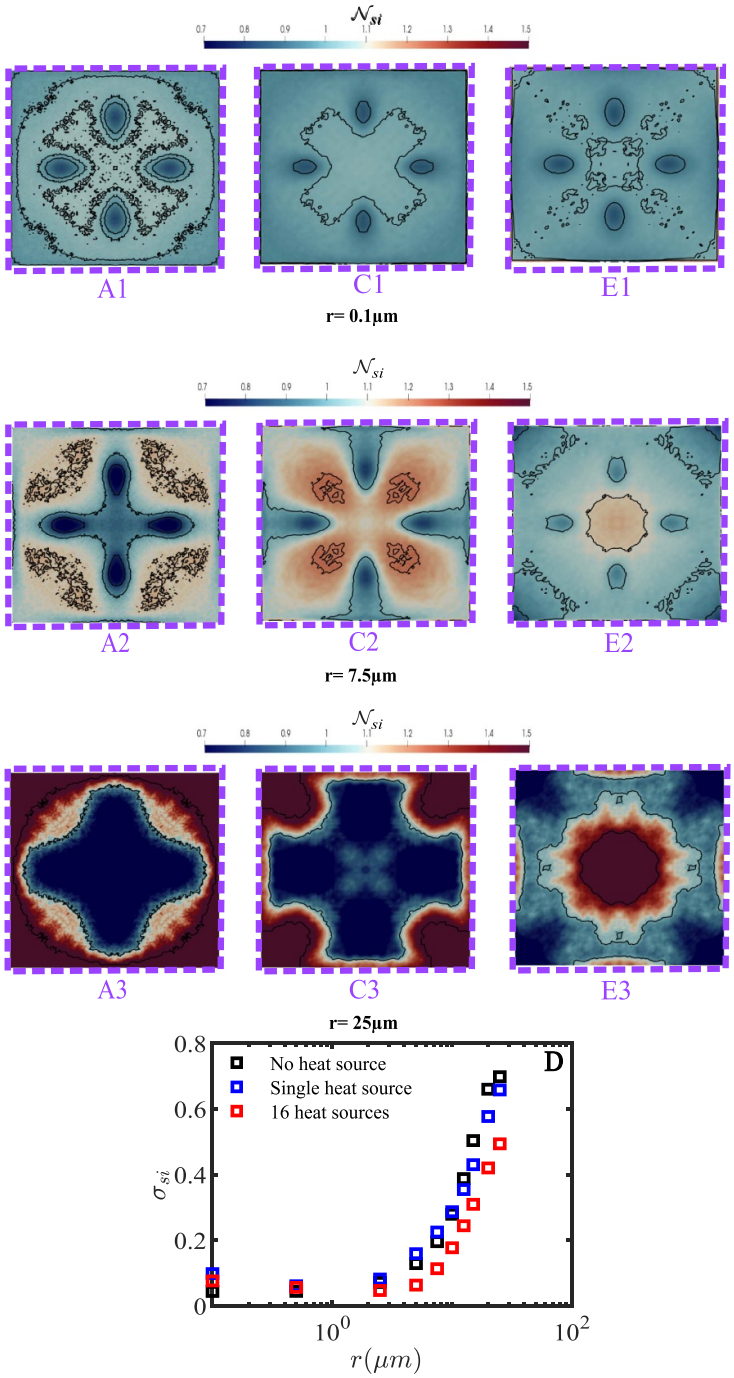


Figure 4. Concentration from the receiver's perspective for ACH = 2.5. Panels (A), (B), and (C) show contours of nuclei concentration $\mathcal{N}_{si}$ for $r = 0.1$, 7.5, and 25 μm at $z = 1.6$ m for the cases with no heat sources, a single heat source and multiple heat sources, respectively. Panel (D) shows the standard deviation of the PDFs of $\mathcal{N}_{si}$ in the entire room as a function of the nuclei radii.

et al 2014). Both these scenarios are identical in the sense that the incoming air through the inlet vents is perfectly clean and devoid of any virus-laden nuclei. In cases where the incoming air is not perfectly clean, the analysis must be repeated by properly accounting for the effects of recycling and filtration. For example in the extreme case of no filtration, one would assume that a sink located in the paths of the inlet jets would not be exposed to low values of concentration, and similarly, a source located below the outlet would not result in a lower steady-state nuclei concentration.

3.4. Statistics for specified source–sink separations

Salinas *et al* (2022) proposed a dimensionless time- and separation-distance-dependent correction function that could be used in conjunction with the well-mixed model. The time-dependent normalized concentration for nuclei of radius r at a specific distance $|\boldsymbol{\xi}| = d$ is defined as $\hat{n}(t, \boldsymbol{\xi}, r)$, where it is recognized that there are many different combinations of source and sink locations that are d apart and these different source–sink combinations are denoted by the variable $\boldsymbol{\xi} = \mathbf{x}_{\mathrm{so}} - \mathbf{x}_{\mathrm{si}}$. Salinas *et al* (2022) simplified the problem substantially, by integrating over all possible source–sink combinations and over all the nuclei sizes weighted by their ejection rate, to define

$$\langle \hat{n} \rangle_d (t, d) = \frac{\int_0^{r_c} \left(\int_{|\boldsymbol{\xi}|=d} \hat{n}\,(t, \boldsymbol{\xi}, r)\, \mathrm{d}\boldsymbol{\xi} \right) V(r)\, Q_{\mathrm{b}}\, n_{\mathrm{so}}\,(r)\, \mathrm{d}r}{\int_0^{r_c} \left(\int_{|\boldsymbol{\xi}|=d} \mathrm{d}\boldsymbol{\xi} \right) V(r)\, Q_{\mathrm{b}}\, n_{\mathrm{so}}\,(r)\, \mathrm{d}r}\,. \tag{9}$$

The upper limit of the integral r_c represents the size of the largest airborne nuclei and its value is taken from Salinas *et al* (2022). In the above, $V(r) = 4\pi r^3/3$ is the volume of nuclei of radius r, Q_{b} is the breathing rate of the source in volume per time, and $Q_{\mathrm{b}}\, n_{\mathrm{so}}(r)\, \mathrm{d}r$ is the number of airborne nuclei of size $r - \mathrm{d}r/2$ to $r + \mathrm{d}r/2$ ejected by the source per unit time. Q_{b} and n_{so} are quantities that are specific to the source (i.e. the sick person) and the nature of the ejection activity, such as breathing or talking. The corresponding size-integrated well-mixed normalized nuclei concentration can be calculated as

$$\langle \hat{n}_{wm} \rangle\,(t) = \frac{\int_0^{r_c} \hat{n}_{wm}\,(t, r)\; V(r)\; Q_{\mathrm{b}}\, n_{\mathrm{so}}\,(r)\, \mathrm{d}r}{\int_0^{r_c} V(r)\; Q_{\mathrm{b}}\, n_{\mathrm{so}}\,(r)\, \mathrm{d}r}\,. \tag{10}$$

The time- and distance-dependent correction function may then be expressed as

$$\langle \hat{n} \rangle_d\,(t, d) = \gamma\,(t, d)\; \langle \hat{n}_{wm} \rangle\,(t)\;. \tag{11}$$

Thus, with the correction function, one can modify the well-mixed model to improve the prediction accuracy when the source–sink separation distance is known. Based on the above definition, the correction function does not depend on the breathing rate Q_{b} (also, $\langle \hat{n} \rangle_d (t, d)$ and $\langle \hat{n}_{wm} \rangle$ do not depend on Q_{b}). The ejection size-distribution n_{so} must however be specified. Here we assume the ejection activity to be mouth breathing and use the experimental results of Morawska for the corresponding $n_{\mathrm{so}}(r)$ (Morawska *et al* 2009). Fortunately, the size-integrated results to be presented do not strongly depend on the nature of the ejection activity.

Here we investigate the effect of internal heating on the correction function γ. In figure 5, we plot the temporal evolution of γ for four different separation distances d and for the expiratory activity of mouth breathing (Morawska *et al* 2009). The γ curves can be used to readily obtain the nuclei concentration in conjunction with the well-mixed prediction. The black, blue, and red squares indicate the steady-state values of γ for the cases with no heat source, a single

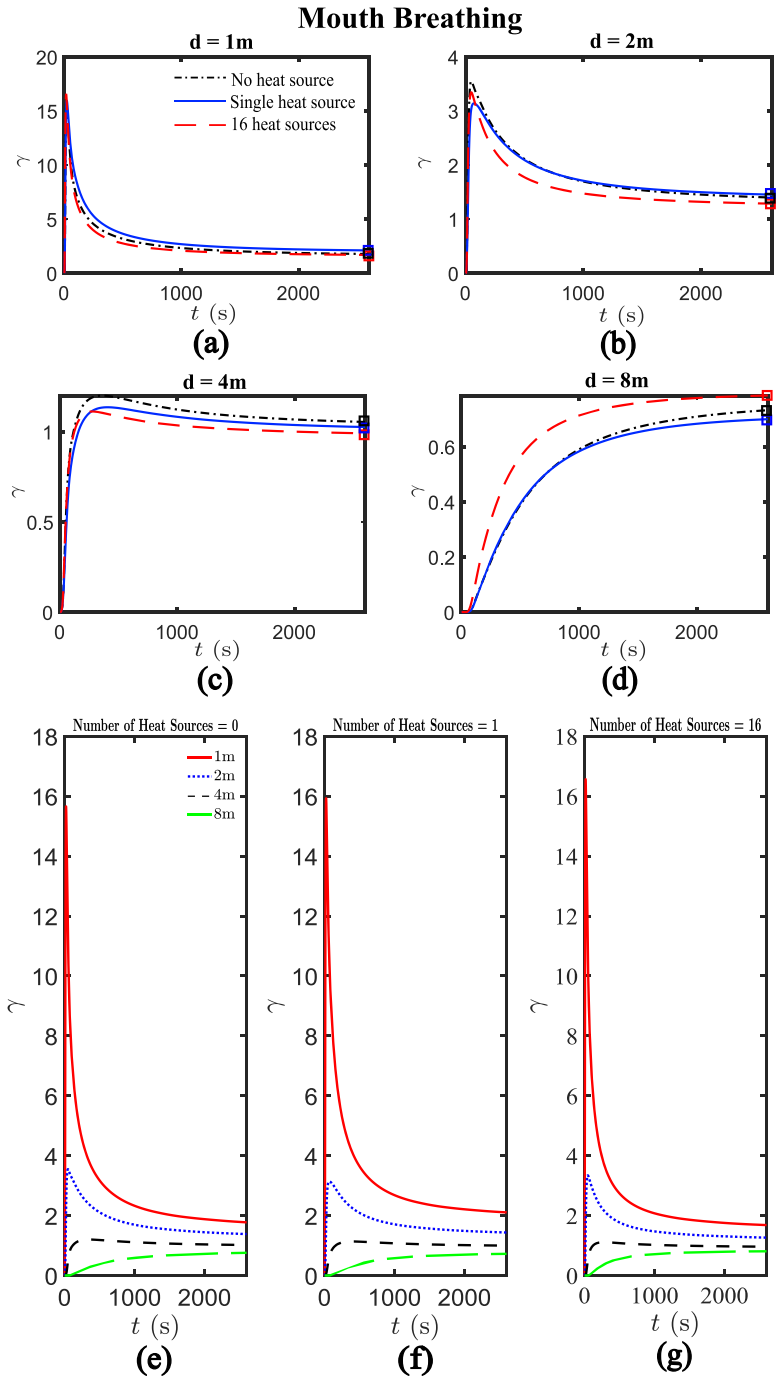


Figure 5. Correction function γ vs time for source–sink separation distances of $d = 1, 2, 4$, and 8 m for the expiratory activity of mouth breathing.The black, blue and red squares indicate the steady-state value of γ for the case of no heat source, single heat source and multiple heat sources respectively. Panels (a)–(d) showcase γ for the three cases for each separation distance, while panels (e)–(g) showcase γ for different separation distances for each case.

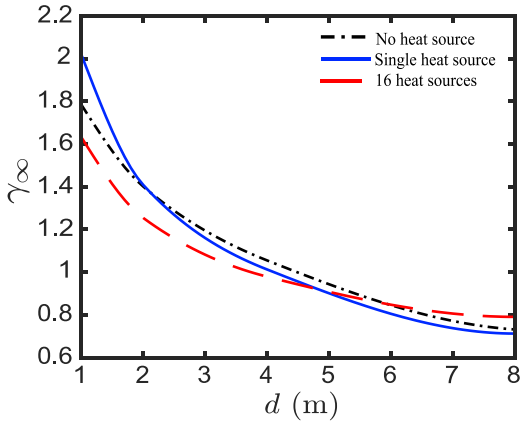


Figure 6. Steady-state correction factor γ_∞ as a function of source-to-sink distance for mouth breathing.

heat source, and multiple heat sources, respectively. For a separation distance of $d = 1$ m, the values are 1.784, 2.019, and 1.632, respectively. At a separation distance of $d = 2$ m, the values are 1.401, 1.409, and 1.254, respectively. For $d = 4$ m, the values are 1.055, 1.013, and 0.979, respectively. Finally, at $d = 8$ m, the values are 0.732, 0.712, and 0.791, respectively. For instance, at a distance of 8 m, and in the absence of any heat sources, γ at $t = 500$ s is about 0.4, indicating that the nuclei concentration is 40% of the well-mixed model prediction as given in (11). According to the figure, the correction function γ is relatively less affected by the presence of a single central heat source compared to the multiple, uniformly distributed sources.

First and foremost, from figure 5, we observe that the presence of heat sources does not greatly alter the overall behavior, namely that at short distances, γ reaches a large peak at short time and then decays to approach a steady-state value, while at large separation distances, γ increases monotonically to reach the steady-state value. A closer examination reveals that multiple uniformly-spaced heat sources tend to enhance mixing. This is evident by the consistently larger value of γ at the larger separation distance of $d = 8$ m. This larger value indicates that nuclei are more likely to propagate over larger distances due to enhanced mixing induced by the distributed heat sources. This increase is balanced by a reduction in nuclei concentration at shorter separation distances. The enhanced mixing can also be quantified by the value of γ and how it generally approaches unity for the case with multiple heat sources compared to the other two cases. A constant value of $\gamma = 1$ for all d indicates that the room is perfectly mixed. In the case of the solitary heat source, its effect, though not strong, generally tends to increase the level of inhomogeneity in the room. This may be due to placing the heat source in the center of the room below the outlet such as its effect remains confined to the central region of the room. The results shown in the figure are for the expiratory activity of mouth breathing. However, it has been shown that the value of γ is largely insensitive to the type of expiratory activity (Salinas *et al* 2022, Krishnaprasad *et al* 2023, Choudhary *et al* 2023a). So, the results of figure 5 remain applicable for any expiratory activity.

In figure 6, we plot the steady-state value $\gamma(t \to \infty, d) = \gamma_\infty(d)$ as a function of the separation distance d. The steady-state value is indicative of the situation where the source has been present for a very long time in the indoor space such that an equilibrium state is achieved where

the rate of ejection by the source is balanced by the rate of removal. Here, we can clearly see how the presence of the multiple heat sources is enhancing mixing by driving the value of γ_∞ towards unity compared to the other two cases. Alternatively, we can also notice that the single centrally-located heating source contributes to the inhomogeneous effect, as evidenced by the heightened variation of γ_∞. It should be noted that the values of γ and γ_∞ in figures 5 and 6 represent an ensemble average of tens of thousands of source–sink combinations separated by the particular distance d.

4. Conclusions

We conducted Euler–Lagrange LES to assess the effect of internal heating on indoor airborne transmission. We considered a canonical indoor space of dimensions $10\,\mathrm{m} \times 10\,\mathrm{m} \times 3.2\,\mathrm{m}$ with a 4-way cassette AC system positioned at the center of the ceiling. The room is initially randomly but uniformly seeded with 10 million droplet nuclei of various sizes that are individually tracked for the duration of the simulations. We conducted two LES simulations whose setup is identical to one another except for the number and position of heat sources within. The first simulation contained a solitary heat source positioned at the center of the room immediately below the ventilation outlet. The second simulation contained 16 equispaced heat sources arranged in a non-staggered arrangement. The heat sources are identical in both simulations and mimic heat generation from an adult person. The heat sources induce a buoyancy-driven flow that influences the trajectories of the droplet nuclei and consequently the spatio-temporal evolution of droplet nuclei concentration. The present results are compared to data from a previous simulation that shares an identical setup and procedure but does not account for any thermal load.

We assess the effect of internal heat sources at multiple levels depending on whether the position of the infected (source) and the susceptible (sink) occupants is known. At the room-average level, we observe the nuclei concentration within the room, which results from a balance of droplet nuclei production through expiratory events and removal through various mechanisms, is largely unaffected by the presence of heat sources, especially for the smallest micron-sized nuclei that are primarily removed from the room through the ventilation outlet. We do however observe that the removal rate of larger nuclei is somewhat enhanced due to the presence of heat sources. On the other hand, the spatial and temporal dispersion of droplet nuclei is significantly influenced by the presence of heat sources when, the source or sink location is known, or when a combination of a source and a sink at a fixed separation distance is considered.

More specifically, we observe that nuclei are better mixed when heat sources are uniformly distributed in the room, and as such, nuclei are more likely to advect over larger distances when the effect of the buoyancy-driven flow is accounted for. Previously, a correction function to the well-mixed model was proposed by Salinas *et al* (2022) to account for the important effect of source–sink spatial separation in the prediction of droplet nuclei concentration. Here, we examine how this correction function γ that is dependent on source-to-sink separation and occupancy time varies with the introduction of the heat sources. In general, in the presence of spatially distributed heat sources, the inhomogeneity of nuclei concentration within the room decreases. On the other hand, localized heat sources may tend to increase the level of inhomogeneity.

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